

CareerX: A Retrieval-Augmented Generation Framework for Personalized AI-Driven Career Guidance

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Abstract—Career guidance systems must adapt to rapidly shifting labor markets, yet traditional platforms rely on static databases while standalone large language models (LLMs) are limited by training data cutoffs and prone to hallucination [9]. This paper presents CareerX, a web-based career guidance system that augments LLM generation with live web retrieval through a SearXNG-based meta-search pipeline. The system dynamically generates personalized intake questionnaires via LLM-driven schema generation, formulates targeted search queries, scrapes and cleans live web sources, and synthesizes structured career dashboards through Zod-enforced schema-constrained generation. Unlike canonical RAG architectures that retrieve from pre-built vector stores, CareerX performs direct web retrieval — live search results are scraped and injected into the LLM context window, trading embedding-based semantic precision for real-time data currency. The system produces location-specific salary data, source-attributed university recommendations, and traceable URL citations that standalone LLMs cannot provide without retrieval augmentation. Initial measurements across three successful test profiles indicate a mean end-to-end pipeline time of 66.7 seconds (SD = 6.2s), of which LLM synthesis accounts for approximately 15–20 seconds, with an average of 37.7 web sources retrieved per session and 79% schema completeness.

Keywords—Retrieval-Augmented Generation, Career Guidance Systems, Generative AI, Web-Augmented Generation, Web Scraping, Educational Technology, Schema-Constrained Generation.

I. INTRODUCTION

A. Problem

In the modern professional landscape, technological disruptions rapidly shift required skill sets, compensation structures, and demanded roles. Conventional career guidance systems rely on static, manually maintained databases such as O*NET [1] that quickly become outdated. When standalone LLMs are used for career counseling, they face two fundamental limitations: training data cutoffs that prevent awareness of

current market conditions, and hallucination of fabricated institutions, salary figures, or career pathways [9].

B. Existing Limitations

Current digital advising tools typically implement rigid, tree-based questionnaires leading to pre-defined outcomes. They cannot dynamically adjust to a user's unique context mid-assessment. When leveraging standalone LLMs without retrieval capabilities, recommendations lack current salary data, emerging Applicant Tracking System (ATS) criteria, and recently established academic programs. Recent systems such as those described by Jawhar et al. [3] have explored AI-powered university and career guidance, but rely primarily on static knowledge bases without real-time data integration.

C. Motivation

The motivation behind this work is to bridge the gap between static counseling and live industry intelligence. By augmenting LLM generation with real-time web retrieval, a system can ground its recommendations in current, verifiable data rather than relying solely on potentially outdated training knowledge.

D. System Overview

CareerX is an end-to-end system that combines adaptive user profiling with real-time web data retrieval and schema-constrained LLM synthesis. The system operates through a multi-stage pipeline: dynamic questionnaire generation, contextual search query formulation, web data aggregation via SearXNG, HTML scraping and cleaning, and structured dashboard synthesis through Zod-validated LLM output. This layered approach enables the delivery of personalized, location-aware career guidance grounded in current web data.

E. Contribution

The main contributions of this work are:

- 1) A dynamic questionnaire generation mechanism that adapts intake questions to user context through LLM-driven schema generation.
- 2) A web-augmented retrieval pipeline that fetches, scrapes, and processes live web sources through SearXNG meta-search aggregation, providing current context for LLM synthesis.
- 3) A schema-constrained synthesis layer using Zod validation to enforce structured, predictable JSON outputs suitable for direct UI rendering.
- 4) Preliminary quantitative evaluation across diverse user profiles demonstrating the system's response characteristics, output completeness, and retrieval volume.

F. Differentiation from Prior Work

CareerX extends the conceptual framework of AI-powered career guidance systems such as [3] in several key ways. While [3] relies on static knowledge and pre-defined recommendation logic, CareerX introduces live web retrieval as a primary data source, dynamic (non-hardcoded) questionnaire generation, and schema-constrained LLM output synthesis. Table III provides a detailed architectural comparison.

II. RELATED WORK

A. Traditional Career Guidance Systems

Career guidance has historically relied on expert systems and static databases. The O*NET framework provides standardized occupational data but requires manual updating and cannot reflect rapid market shifts [1]. Framework designs for intelligent career recommendation systems for undergraduates have been explored by Liu et al. [1], and AI applications in education and career services have been studied across multiple domains [2], [6]. While these approaches provide foundational value, they are inherently limited by the currency of their underlying data.

B. Large Language Models in Guidance

The emergence of large language models has prompted their application in advisory domains. Models such as GPT-4 have demonstrated significant improvements in factual reasoning established the foundation for prompt-based advisory systems, as explored by Brown et al. [4]. Yet all standalone LLMs face an inherent limitation: they cannot access information beyond their training cutoff date. Ji et al. [8] provide a comprehensive survey of hallucination in natural language generation, documenting how LLMs frequently generate plausible but fabricated information — a particularly dangerous failure mode in career guidance where users may make life decisions based on recommended institutions or salary expectations.

C. Retrieval-Augmented Generation

Lewis et al. [6] introduced Retrieval-Augmented Generation (RAG), demonstrating that grounding LLM outputs in retrieved documents significantly reduces hallucination and improves factual accuracy for knowledge-intensive tasks. The canonical RAG architecture retrieves documents from a vector store using embedding-based similarity search. Subsequent work by Asai et al. [9] on Self-RAG introduced adaptive retrieval where the model decides when retrieval is necessary. Jiang et al. [10] proposed FLARE (Forward-Looking Active Retrieval), which anticipates future information needs during generation. Commercial systems such as Perplexity.ai and You.com have demonstrated the viability of real-time web retrieval augmented generation at scale. However, these systems are general-purpose and not optimized for structured domain-specific outputs with schema-constrained rendering.

CareerX differs from canonical RAG architectures in its retrieval mechanism: rather than querying a pre-built vector store, it performs live web searches and scrapes results in real-time. This trades embedding-based semantic precision for data currency — the system always operates on the most recent publicly available information.

D. Gap Analysis

Existing approaches lack the integration of three capabilities simultaneously: (1) real-time web retrieval as a primary knowledge source, (2) adaptive user modeling through dynamic questionnaire generation, and (3) schema-constrained output synthesis for reliable UI rendering. Systems such as Perplexity.ai address capability (1) and partially (3) through structured answer formatting, but lack adaptive intake modeling — they operate on single-turn queries rather than multi-stage profiling. Traditional career guidance systems such as O*NET address domain specificity but lack capabilities (1) and (2). CareerX is designed to integrate all three within a career guidance context. Risk-aware deployment considerations for AI systems, as emphasized by NIST [7], are incorporated through the system's privacy-preserving architecture design.

III. SYSTEM ARCHITECTURE

The system architecture comprises three primary components:

A. Client Interface

Built on Next.js 15 and Framer Motion, the client renders dynamic UI components based on AI-generated JSON schemas. The questionnaire interface adapts to the LLM's output schema, supporting multiple input types (multi-select, sliders, rating scales, text fields). The dashboard visualizes career pathways as interactive nodes with match percentages, salary ranges, and source citation links.

B. AI Orchestration Layer

Powered by Google Gemini 1.5 Flash, this layer performs three distinct generation tasks: (1) intake questionnaire schema generation based on user role and context, (2) search query

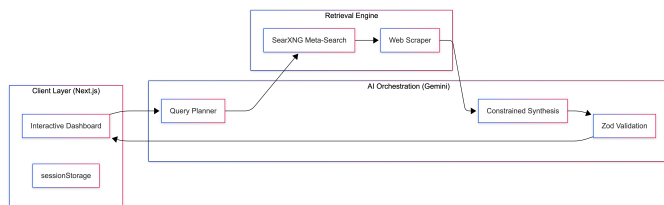


Fig. 1. CareerX system architecture illustrating the three-layer pipeline: client interface, AI orchestration, and data retrieval engine.

planning to formulate 4-8 targeted web queries, and (3) dashboard data synthesis from retrieved web content. All LLM outputs are validated against Zod schemas to enforce structural correctness and prevent malformed responses from reaching the client.

C. Data Retrieval Engine

A self-hosted SearXNG meta-search instance aggregates results across multiple search engines. The system generates 4-8 targeted search queries per user session, retrieves up to 5 pages per query (with 1.2-second inter-page delays), and deduplicates results by URL. A custom web scraper with 12-second timeout handling and graceful degradation on inaccessible sources (HTTP 403/404) extracts clean text content from the top 10 results. Each scraped page is limited to 10,000 characters after HTML tag, script, and stylesheet removal to manage LLM context window utilization. A banned domain list filters out dictionary and definition sites that do not contribute career-relevant content.

D. Pipeline Overview

The system executes a five-stage pipeline: (1) User intake — name, role, location, and geolocation are captured; (2) Dynamic questionnaire — the LLM generates an adaptive JSON schema of 5-7 questions, rendered as interactive UI components; (3) Query planning and retrieval — the LLM formulates 4-8 search queries, SearXNG retrieves and paginates results, and the top 10 URLs are scraped for full content; (4) Context assembly — scraped content (~10,000 chars each) plus 50 additional result snippets are combined into a context payload of approximately 80,000-100,000 characters; (5) Schema-constrained synthesis — the LLM generates a Zod-validated career dashboard from the assembled context.

IV. METHODOLOGY

The following subsections detail the technical implementation of pipeline stages 3–5 described in Section III-D.

A. Retrieval Strategy

The system does not rely on a predefined static dataset for recommendations, but instead utilizes real-time web data as its primary knowledge source. When a user begins a session, CareerX identifies their primary axis (role, education level) and geolocation (via IP-API with manual fallback). The LLM generates 4-8 search queries covering: role-specific salary data, community opinions (targeting Reddit and LinkedIn via

site-specific queries), university programs, certifications, and location-specific opportunities.

B. Context Assembly

The metasearch pipeline retrieves URLs across up to 5 pages per query, yielding 30-60 unique results per session after URL deduplication. The top 10 results are scraped for full page content (up to 10,000 characters each after HTML/CSS/script tag removal), while an additional 50 results contribute snippet-level context (typically 200-500 characters each from SearXNG result excerpts). This combined context is provided to the synthesis LLM alongside the user's questionnaire responses.

C. Schema-Constrained Synthesis

The synthesis LLM generates output conforming to a comprehensive Zod schema comprising: career pathways (with match percentages, salary ranges, and market demand indicators), education pathways (degrees + certifications with provider URLs), ranked universities (with fees and program listings), job mappings (with portal deep-links to LinkedIn, Naukri, Indeed), skill gap analysis (with year-by-year roadmaps), and source citations mapping to retrieved URLs.

The match percentages in career pathways are computed internally by the LLM based on its assessment of alignment between user attributes and the retrieved market data. Schema-constrained generation approaches have been shown to significantly improve the reliability and parsability of LLM outputs in downstream applications [12], [13].

V. IMPLEMENTATION

A. Technology Stack

Frontend: Next.js 15 (App Router), TailwindCSS, Framer Motion. Backend: Next.js serverless API routes (Node.js runtime). AI Core: Google Gemini 1.5 Flash via the Vercel AI SDK. Search Infrastructure: SearXNG (self-hosted Docker instance). Schema Validation: Zod.

B. System Integration

The platform uses in-session client-side state (React component state and sessionStorage) rather than persistent server-side databases. User data is processed within the active browser session and is not retained on the server after response delivery. No user data is written to server-side storage at any point in the pipeline. sessionStorage is automatically cleared when the browser tab is closed.

VI. EVALUATION

A. Experimental Setup

The system was evaluated using five simulated user profiles (P1–P5) representing diverse backgrounds. Testing was conducted on a MacBook with Apple Silicon, 16GB RAM, and a residential broadband connection (~50 Mbps), using the Gemini API free tier (rate limit: 20 requests/minute).

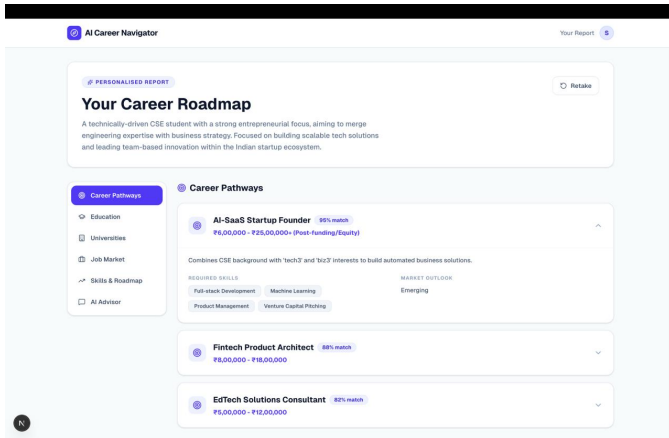


Fig. 2. CareerX dashboard interface showing interactive career pathway nodes, salary data, and source citations.

B. Schema Completeness Metric

Schema completeness is defined as:

$$Completeness = \frac{N_{populated}}{N_{total}} \times 100 \quad (1)$$

where $N_{total} = 11$ (6 required + 5 optional fields).

C. Quantitative Results

Of the five profiles tested, three completed successfully; two were terminated due to API rate limiting. Table II presents the per-profile results.

TABLE I
DETAILED FEATURE COMPARISON

Feature	O*NET	Standalone LLM	CareerX
Dynamic Questionnaire	No	Partial	Yes
Live Web Retrieval	No	No	Yes
Schema-Constrained	Yes	No	Yes
Source Citations	Yes	No	Yes
State-less Privacy	No	Partial	Yes

TABLE II
PRELIMINARY EVALUATION RESULTS

Profile	Time (s)	Sources	Paths	Unis	Jobs	Cmp%
P1 (India)	68.4	37	2	3	2	82%
P2 (USA)	73.3	34	3	3	2	82%
P3 (UK)	58.3	42	2	3	2	73%
P4	—	—	—	—	—	—
P5	—	—	—	—	—	—

Preliminary aggregate metrics (n=3): Mean end-to-end response time = 66.7s (SD = 6.2s). Mean sources retrieved = 37.7 per session. Mean schema completeness = 79.0%.

D. Architectural Comparison

Table III contrasts CareerX with traditional systems.

TABLE III
ARCHITECTURAL COMPARISON

Feature	O*NET	Standalone LLM	CareerX
Data Source	Static	Training Data	Live Web + Training
Retrieval	DB Query	None	Web Search
Output	Templates	Unstructured	Schema JSON
Currency	Manual	Cutoff	Real-time

VII. LIMITATIONS

1) Sample Size: Preliminary evaluation (n=3) is indicative only. 2) Retrieval Quality: Systematic precision analysis is needed. 3) Retrieval Depth: Trades embedding precision for currency. 4) Match Percentage Opacity: Scores are LLM-generated qualitative indicators. 5) Hallucination Risk: Grounded context reduces but does not eliminate risk. 6) No User Study: Evaluation uses simulated profiles. 7) Scraping Ethics: Requires robots.txt parsing implementation.

VIII. CONCLUSION

This paper presents CareerX, a system that augments LLM-based career guidance with live web retrieval. Preliminary evaluation demonstrates that the system retrieves an average of 37.7 sources per session and produces schema-constrained outputs with 79% completeness. The architecture demonstrates the practical viability of combining live web retrieval with schema-constrained generative AI for domain-specific advisory applications.

FUTURE WORK

1) Expanding the evaluation to $n \geq 10$ profiles. 2) Conducting retrieval precision analysis via manual annotation. 3) Implementing a formal user study (n=20–30). 4) Adding embedding-based retrieval for semantic re-ranking. 5) Conducting formal hallucination evaluation via RAGAS. 6) Implementing an ablation study. 7) Optimizing response time through parallel query execution.

REFERENCES

- [1] W. Liu, L. Niu and J. Zhao, "Framework Design of Intelligent Career Recommendation System for Undergraduates," 2022 International Conference on Intelligent Education and Intelligent Research (IEIR), Wuhan, China, 2022, pp. 256-261, doi: 10.1109/IEIR56323.2022.10050076.
- [2] F. Yan, Y. Liu, J. Liang and Z. Li, "Academic Planning and Career Choice of Art Doctoral Students in the Era of Artificial Intelligence," 2019 International Joint Conference on Information Media and Engineering (IJCIME), pp. 388-390, 2019.
- [3] R. Petkov, "Digital Media Literacy, Artificial Intelligence and Modernization of Youth Career Services," 2021 XXX International Scientific Conference Electronics (ET), Sozopol, Bulgaria, 2021, pp. 1-4, doi: 10.1109/ET52713.2021.9579955.
- [4] T. Brown et al., "Language Models are Few-Shot Learners," Advances in Neural Information Processing Systems (NeurIPS), 2020.
- [5] M. Neumann, M. Rauschenberger and E.-M. Schon, "We Need to Talk About ChatGPT: The Future of AI and Higher Education," 2023 IEEE/ACM SEENG Workshop, pp. 29-32, 2023.
- [6] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems (NeurIPS), 2020.

- [7] National Institute of Standards and Technology (NIST), "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," 2023. [Online]. Available: <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>
- [8] Z. Ji et al., "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1-38, 2023, doi: 10.1145/3571730.
- [9] A. Asai et al., "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection," *arXiv:2310.11511*, 2023.
- [10] Z. Jiang et al., "Active Retrieval Augmented Generation," *arXiv:2305.06983*, 2023.
- [11] S. Es et al., "RAGAS: Automated Evaluation of Retrieval Augmented Generation," *arXiv:2309.15217*, 2023.
- [12] S. Gupta, R. Ranjan and S. N. Singh, "A Comprehensive Survey of Retrieval-Augmented Generation (RAG): Evolution, Current Landscape and Future Directions," *arXiv:2410.12837*, 2024.
- [13] S. Geng et al., "JSONSchemaBench: A Rigorous Benchmark of Structured Outputs for Language Models," *arXiv:2501.10868*, 2025.
- [14] Y. Siriwardhana et al., "Retrieval-Augmented Generation for Educational Applications: A Systematic Survey," *Computers and Education: Artificial Intelligence*, vol. 10, 2025, doi: 10.1016/j.caeai.2025.100578.
- [15] Y. Du et al., "Enhancing Job Recommendation through LLM-Based Generative Adversarial Networks," *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI-24)*, pp. 8363-8371, 2024.
- [16] C. R. Kishore and T. Sreerala, "Personalized Job Search with AI: A Recommendation System Integrating Real-Time Data and Skill-Based Matching," *International Journal of Engineering Research and Technology*, vol. 14, no. 4, 2025.
- [17] R. R. Mirajkar et al., "Skill Recommendation System and Resume Analysis using AI," *2024 IEEE Pune Section International Conference (PuneCon)*, Pune, India, 2024, pp. 1-3, doi: 10.1109/PuneCon63413.2024.10895306.
- [18] M. X. Liu et al., "We Need Structured Output: Towards User-Centered Constraints on Large Language Model Output," *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1-9, 2024, doi: 10.1145/3613905.3650756.
- [19] A. Neumann et al., "Retrieval-Augmented Generation (RAG) Chatbots for Education: A Survey of Applications," *Applied Sciences*, vol. 15, no. 8, p. 4234, 2025, doi: 10.3390/app15084234.